

CO2 AT3 DEBUGGING & OPTIMIZATION TASK

Course Code: ITA0614 - Machine Learning

Name: G Sathya

Reg.No: 192321094

1. A decision tree model trained on a large dataset shows high accuracy on training data but poor performance on unseen data. Analyze possible causes such as overfitting, deep tree structure, or noisy data. Propose debugging steps and optimization methods like pruning, cross-validation, and feature selection to improve generalization.

Solution:

A decision tree model performs very well on the training dataset but poorly on unseen data. This indicates that the model has learned the training data too closely and cannot generalize effectively. This problem is commonly known as overfitting.

Possible Causes

1. Overfitting

- The tree memorizes training samples instead of learning patterns.
- Achieves nearly 100% training accuracy but low testing accuracy.
- Poor generalization to new data.

2. Deep Tree Structure

- Excessive depth creates many branches.
- Learns minor variations and noise in data.
- Results in high variance.

3. Noisy Data

- Dataset may contain incorrect or inconsistent values.
- Tree creates unnecessary splits to fit noise.
- Leads to poor predictive performance.

4. Irrelevant Features

- Unimportant attributes may influence decision making.
- Increases model complexity.
- Reduces accuracy on unseen data.

Debugging Steps

Step 1: Compare Training and Testing Accuracy

Example:

Dataset	Accuracy
Training	98%
Testing	72%

Large difference indicates overfitting.

Step 2: Visualize the Tree

- Check tree depth.
- Identify unnecessary branches.
- Detect highly complex structures.

Step 3: Analyze Feature Importance

- Remove features contributing very little.
- Focus on significant predictors.

Step 4: Examine Data Quality

- Check missing values.
- Remove duplicates.
- Handle outliers and noisy records.

Step 5: Use Validation Data

- Evaluate performance on unseen validation samples.
- Monitor model behavior before deployment.

Optimization Methods

1. Pruning

Pruning removes unnecessary branches.

Pre-Pruning

- Limit maximum depth.
- Set minimum samples per leaf.
- Stop splitting early.

Post-Pruning

- Grow complete tree first.
- Remove branches that do not improve accuracy.

Benefits:

- Reduces complexity.
- Improves generalization.

2. Cross-Validation

K-Fold Cross Validation:

1. Divide dataset into K parts.
2. Train on K-1 parts.
3. Test on remaining part.
4. Repeat K times.

Benefits:

- More reliable performance estimate.
- Detects overfitting.

3. Feature Selection

Methods:

- Information Gain
- Chi-Square Test
- Correlation Analysis

Benefits:

- Removes irrelevant features.
- Simplifies tree structure.
- Improves prediction accuracy.

4. Hyperparameter Tuning

Adjust:

- Maximum depth
- Minimum samples split
- Minimum samples leaf

Benefits:

- Controls tree growth.
- Prevents overfitting.

5. Ensemble Methods

Use:

- Random Forest
- Gradient Boosting

Benefits:

- Better generalization.
- Higher accuracy.

2. A convolutional neural network (CNN) used for image classification produces unstable results across training epochs. Examine potential issues such as improper data augmentation, vanishing gradients, or batch normalization errors. Describe debugging strategies and suggest optimization techniques to stabilize training and improve accuracy.

Solution:

A Convolutional Neural Network (CNN) is used for image classification, but its accuracy and loss fluctuate significantly during training. Such instability indicates issues in data processing, network design, or training configuration.

Possible Causes

1. Improper Data Augmentation

Examples:

- Excessive image rotation.
- Random cropping removing important objects.
- Unrealistic transformations.

Effects:

- Model receives inconsistent inputs.
- Training becomes unstable.

2. Vanishing Gradients

Occurs when:

- Network is very deep.
- Gradients become extremely small during backpropagation.

Effects:

- Early layers learn very slowly.
- Accuracy fluctuates.

3. Batch Normalization Errors

Possible issues:

- Incorrect placement of BatchNorm layers.
- Very small batch size.
- Improper running statistics.

Effects:

- Inconsistent feature distributions.
- Unstable training behavior.

4. High Learning Rate

When learning rate is too large:

- Weights change drastically.
- Loss oscillates instead of converging.

Effects:

- Accuracy jumps up and down.

5. Insufficient or Imbalanced Dataset

Problems:

- Few training images.

- Certain classes dominate.

Effects:

- Unstable predictions.
- Poor generalization.

6. Overfitting

Symptoms:

- Training accuracy increases.
- Validation accuracy decreases.

Effects:

- Poor performance on new images.

Debugging Strategies

Step 1: Monitor Training and Validation Curves

Observe:

- Training Loss
- Validation Loss
- Training Accuracy
- Validation Accuracy

Purpose:

- Identify overfitting or underfitting.

Step 2: Check Data Augmentation Pipeline

Verify:

- Rotation range
- Cropping size
- Flipping operations

Ensure transformations are realistic.

Step 3: Inspect Gradient Values

Monitor gradients during training.

If gradients:

- Become extremely small → Vanishing Gradient.
- Become extremely large → Exploding Gradient.

Step 4: Verify Batch Normalization

Check:

- BatchNorm placement after convolution layers.
- Batch size is sufficiently large.

Step 5: Experiment with Learning Rate

Try:

- 0.1
- 0.01
- 0.001

Select the most stable value.

Optimization Techniques

1. Use Batch Normalization

Advantages:

- Stabilizes activations.
- Speeds up convergence.
- Improves accuracy.

2. Apply Proper Data Augmentation

Examples:

- Horizontal Flip
- Small Rotations

- Brightness Adjustment

Benefits:

- Increases dataset diversity.
- Reduces overfitting.

3. Learning Rate Scheduling

Reduce learning rate gradually during training.

Methods:

- Step Decay
- Exponential Decay
- Cosine Annealing

Benefits:

- Smooth convergence.

4. Use ReLU Activation

Advantages:

- Reduces vanishing gradient problem.
- Faster training.

5. Add Dropout Layers

Example:

- Dropout Rate = 0.5

Benefits:

- Prevents overfitting.
- Improves generalization.

6. Early Stopping

Stop training when validation loss stops improving.

Benefits:

- Avoids overfitting.
- Saves computation time.

7. Transfer Learning

Use pretrained models such as:

- ResNet
- VGG16
- MobileNet

Benefits:

- Better feature extraction.
- Faster convergence.